

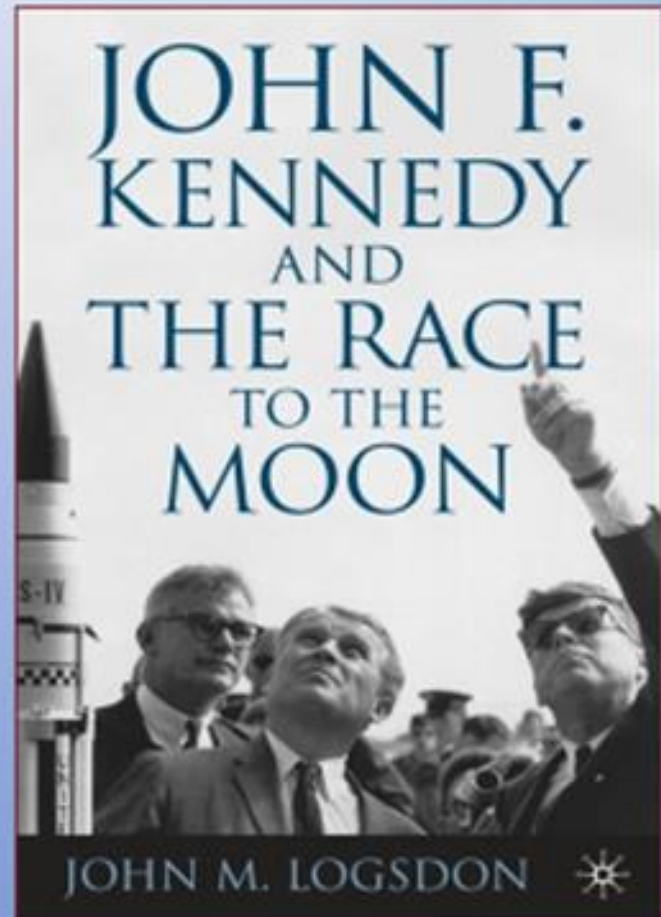
ONCE, WE WENT TO THE MOON

John M. Logsdon
September 7, 2023

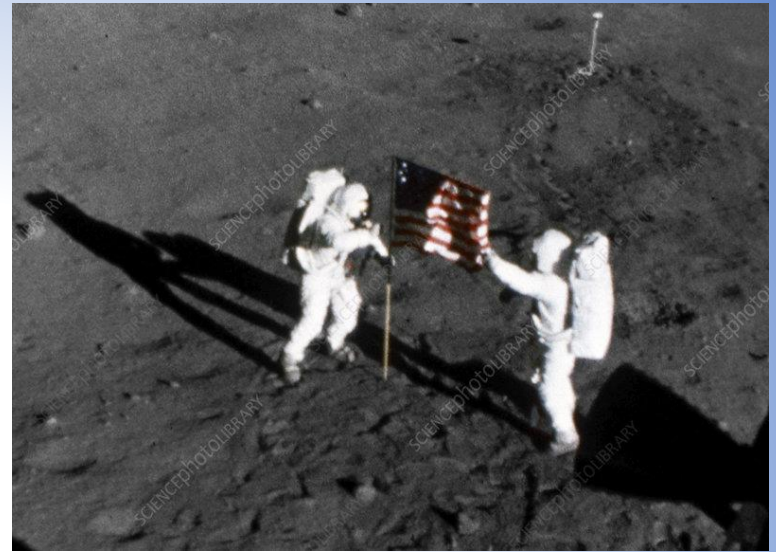
1970



2010



- **Between December 1968 and December 1972, the United States sent nine Apollo missions to the Moon, each carrying three American astronauts**
- **Six of those missions landed on the Moon, and twelve Americans walked on the lunar surface**
- **Apollo 17 left the Moon on December 14, 1972, and no humans have returned since.**



Apollo 11 crew plants U.S. flag on Moon



THREE QUESTIONS

- **Why did the United States go to the Moon?**
- **Why did it stop?**
- **What were the impacts of Apollo?**

ORIGINS OF APOLLO

- **NASA in 1959 chose a lunar landing after 1970 as its long-range goal for human spaceflight**
- **NASA in August 1960 announced its proposal for next steps in human spaceflight**
 - **3-person spacecraft for longer duration orbital flight or flights around the Moon**
 - **This proposed program was named Apollo**
- **NASA in October 1960 also began preliminary planning for a lunar landing mission**
- **President Dwight D. Eisenhower, about to leave the White House, was briefed on NASA's planning in December 1960 and reacted quite negatively**

AS JOHN F. KENNEDY BECAME PRESIDENT

- **“Of all the major problems facing Kennedy when he came into office, he probably knew and understood least about space.” Hugh Sidey, *John F. Kennedy, President* (1964)**
- **JFK initially delegated lead responsibility for space issues in his administration to Vice President Lyndon B. Johnson**
- **At the end of March 1961, Kennedy deferred until Fall 1961 a decision on the future of the U.S. human spaceflight program**

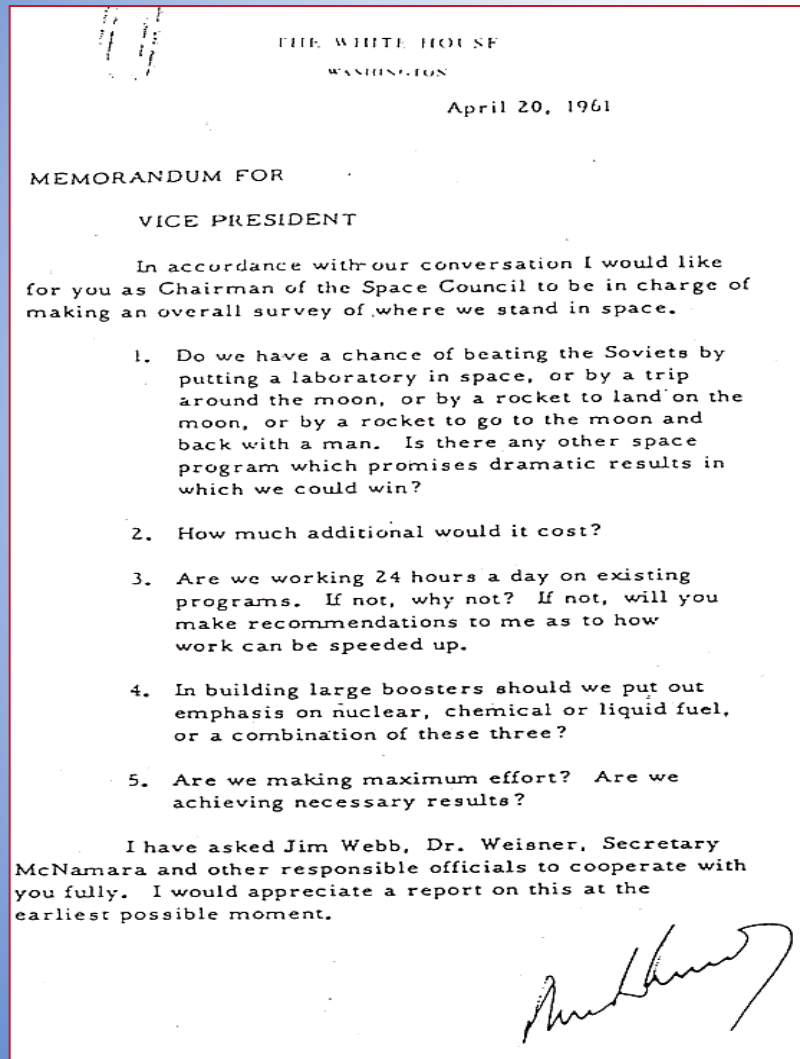
YURI GAGARIN – FIRST HUMAN IN SPACE

APRIL 12, 1961



- As had been the case after *Sputnik* in 1957, the Soviet Union by being first to send a human into orbit gained much positive reaction around the world
- The U.S. domestic reaction to the Soviet success was that it represented another U.S. failure, just as JFK was trying to “get the country moving again.”
- John F. Kennedy in response decided that the United States had to enter – and win – a space race. On April 14, he told his space advisers “There’s nothing more important.”

KENNEDY SETS OUT THE REQUIREMENTS



“Do we have a chance of beating the Soviets by putting a laboratory in space, or by a trip around the moon, or by a rocket to go to the moon and back with a man? Is there any other space program which promises dramatic results in which we could win?”

**John F. Kennedy to Lyndon B. Johnson
April 20, 1961**

WHY THE MOON?

Based on NASA's early planning, NASA leadership had told JFK on April 14 that the U.S. might win a race to the Moon, if adequate new funds were committed.

After JFK's April 20 memo, rocket engineer and visionary Wernher von Braun was consulted; tells Lyndon Johnson that "a rocket about ten times as powerful as the Soviet . . . rocket is required" and "we have an excellent chance of beating the Soviets to the first landing of a crew on the moon."

Getting to the Moon in a sense became a rocket-building race.

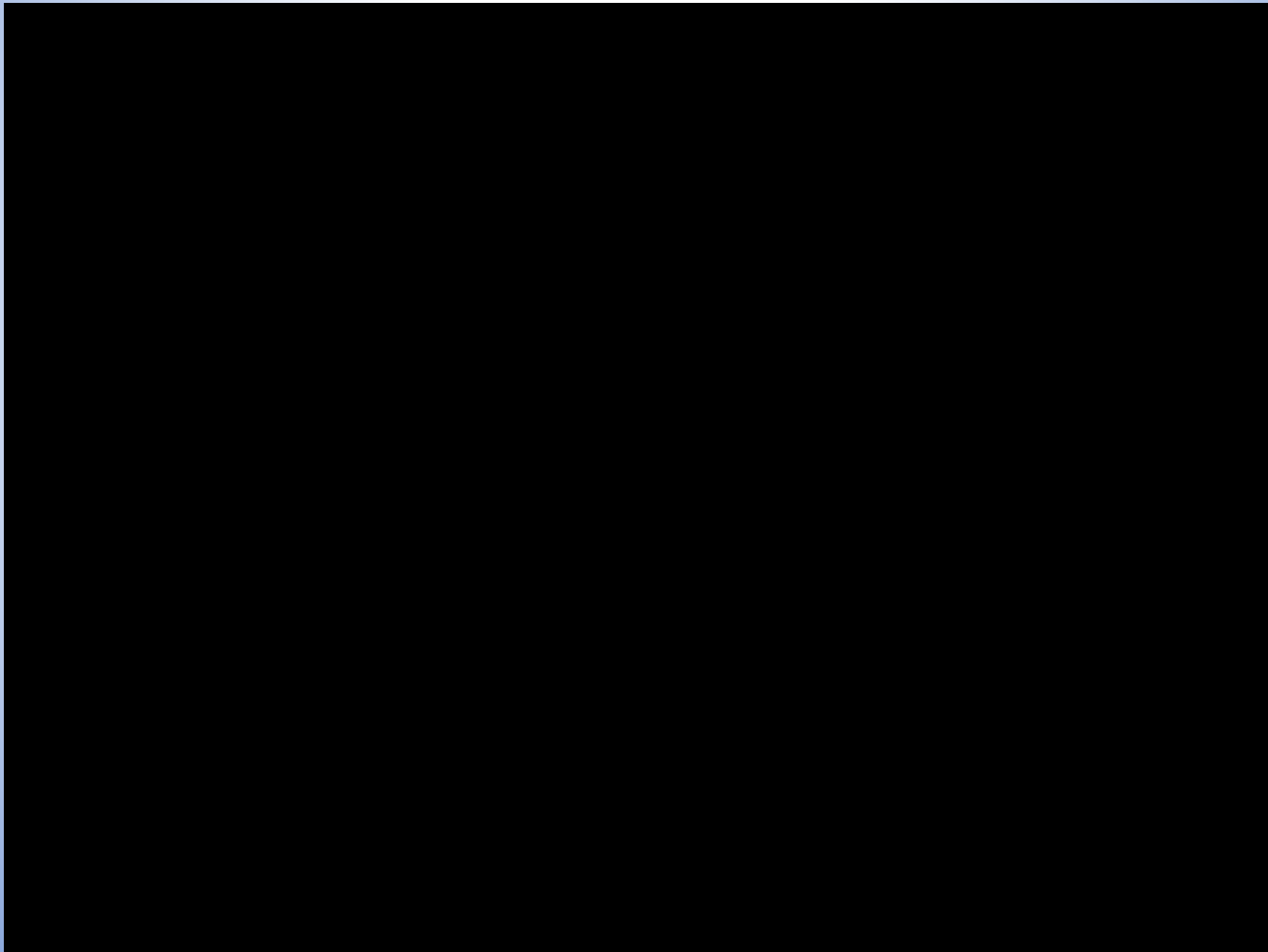
NASA Administrator James Webb and Secretary of Defense Robert McNamara in a May 8, 1961, memo observed

- "It is men, not merely machines, that capture the imagination of the world."
- Large-scale space projects aimed at "enhancing national prestige" are "part of the battle along the fluid front of the cold war."

On May 10 JFK approved their recommendation to set a lunar landing as a national goal



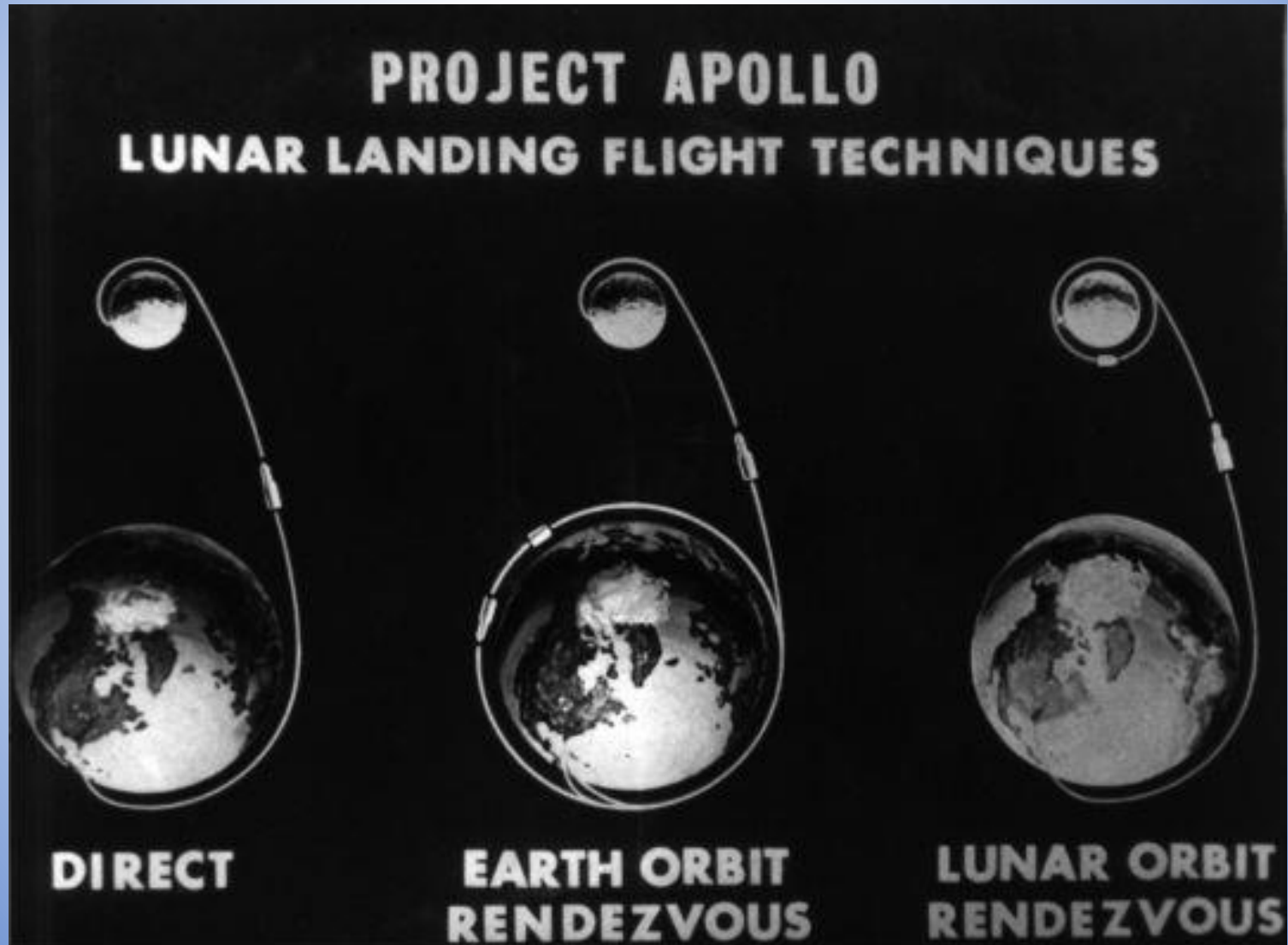
“WE SHOULD GO TO THE MOON”
May 25, 1961



THE APOLLO BUILDUP

- NASA Budget
 - FY1961: \$0.964 billion
 - FY1962: \$1.825 billion – 89% increase
 - FY 1963: \$3.674 billion – 101% increase
 - FY1964: \$5.1 billion – 40% increase
- Construction of facilities
 - FY 1961: \$98.2 million
 - FY 1962: \$217.1 million – 121% increase
 - FY 1963: \$569.8 million – 162% increase
- Personnel
 - 1961: 17, 500 in-house, 57,000 contractors
 - 1962: 23,700 in-house, 115,500 contractors
 - 1963: 29,900 in-house, 218,400 contractors

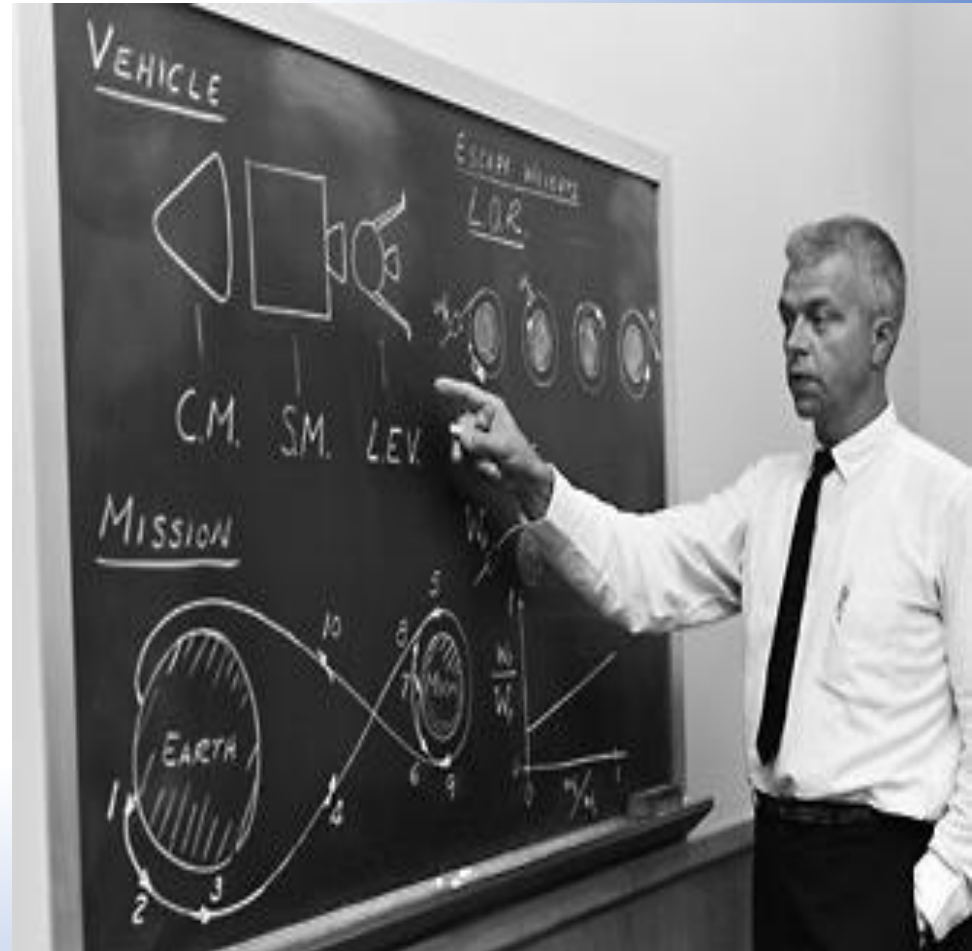
How to Get there?



A KEY DECISION

HOW TO GET TO THE MOON

- There was extensive debate within NASA from May 1961-June 1962 on the best approach to a lunar landing mission
 - Direct ascent
 - Earth orbital rendezvous
 - Lunar orbital rendezvous
- Saturn V launcher selected before this debate was settled
- Wernher von Braun announced his support of LOR in June 1962



John Houbolt

LOR APPROACH DEBATED

- White House science office objected to choice of LOR
 - Did not trust NASA's quantitative analyses
 - Wanted to develop various capabilities in Earth orbit for national security purposes
- Webb and science adviser Wiesner argued as JFK toured Huntsville on September 11, 1962
- In midst of Cuban Missile Crisis, JFK in October 1962 decided to accept NASA's choice



JFK INSPECTS APOLLO'S PROGRESS

- Kennedy visited NASA's Apollo facilities in September 1962 in anticipation of tough budget decisions with respect to funding Apollo
- On September 12 at Rice University in Houston, he made a stirring speech in support of the program

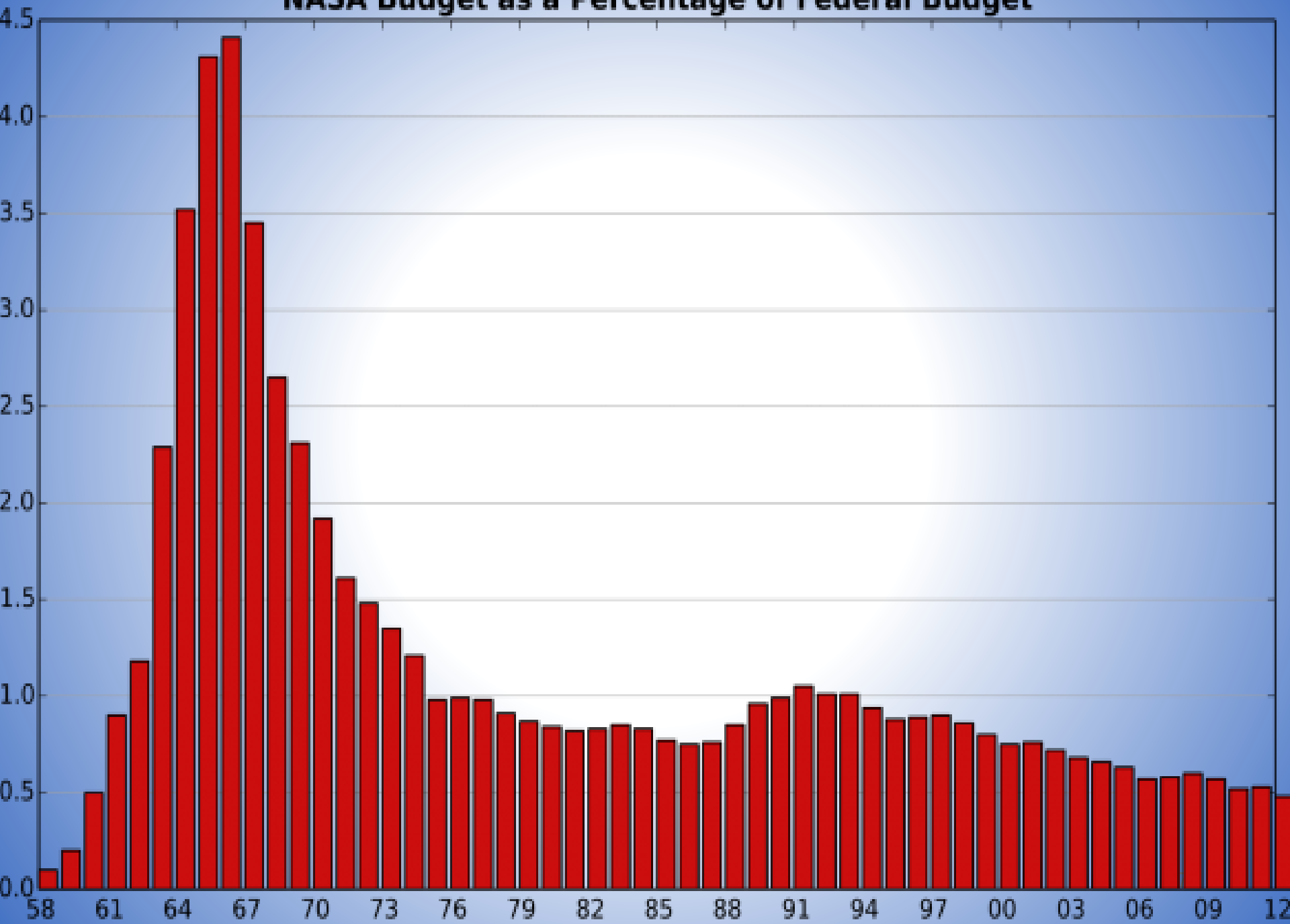


THE COMMITMENT TO APOLLO

- The Apollo buildup was a warlike, but peaceful, mobilization of financial and human resources.
- At its peak in 1966, the NASA budget was 4.4% of the overall government budget. (In 2022, it was 0.5%)
- This commitment was based on the link in JFK's view between space achievements and the Cold War geopolitical competition with the Soviet Union for a global leadership role
- JFK in November 1962 – “the Soviet Union has made this [space leadership] a test of the system . . . Everything that we do ought to really be tied into getting onto the Moon ahead of the Russians.”



NASA Budget as a Percentage of Federal Budget



1963 - GROWING CRITICISMS OF APOLLO

- **Former President Eisenhower, security-oriented Republicans, and influential segments of the scientific and liberal communities were critical of Apollo's priority, each for a different reason**
- **JFK questioned his commitment and asked for reviews of Apollo's benefits, costs, and schedule in both April and August 1963**
- **In August 1963, Congress prepared to cut NASA's budget by over 10%, from requested \$5.7 billion to \$5.1 billion, threatening the "end of decade" target**

WERE CHANGES IN THE WIND?

- **Kennedy in September 1963 ordered a sweeping review of all U.S. space efforts, civilian and military.**
- **During this review, slowing down or even canceling Apollo was considered; some senior JFK advisors favored at least relaxing the “end of the decade” deadline.**
- **Kennedy, aware of flagging public support, proposed giving Apollo a more explicit military justification.**
- **As an alternative, he proposed changing Apollo from a competitive to a cooperative effort**

GOING TO THE MOON WITH THE SOVIET UNION: PURSUIT OF AN ILLUSION?

- In his Inaugural Address, JFK had suggested to the Soviet Union “together let us explore the stars.” He never lost the hope for U.S.-Soviet space cooperation, even as he decided he had to compete
- Ten days after announcing his decision to go to the Moon, JFK met Nikita Khrushchev in Vienna and proposed going to the Moon together. Khrushchev said “no”
- On September 20, 1963, Kennedy spoke to the General Assembly of the United Nations and proposed US – USSR cooperation in lunar exploration
- What might have happened If Khrushchev had said “yes” to JFK’s proposal and Kennedy had lived?



WAS THERE REALLY A SPACE RACE?

- At the time Kennedy made the decision to go to the Moon, the United States did not know whether the USSR had a lunar landing program
- Repeated intelligence estimates in 1962 and 1963 found no hard evidence of a Soviet lunar landing program, but suggested that one was likely to exist
- Soviet Union in mid-1963 publicly said it was not planning a lunar landing program. In fact, such a program was not approved until August 1964, only after extensive internal debate.
- From then on, Soviet Union tried to be first to reach the Moon and to land on the Moon



Launch of a Soviet N-1
Moon rocket, July 3, 1969
(launch failed)

JFK REGAINS HIS ENTHUSIASM

- JFK visited Cape Canaveral on November 16, 1963
- There, he was briefed by head of human spaceflight George Mueller on Apollo plans and preparations.
- Kennedy also was briefed by Wernher von Braun in front of a Saturn 1 booster being readied for a December launch; that launch would give the U.S. the lead in lifting power
- Kennedy came away from his visit very enthused about what he had seen and ready to give his personal support to Apollo.
- Six days later he fell victim to an assassin's bullet



WHAT IF JOHN F. KENNEDY HAD LIVED?

- Kennedy was not a visionary in terms of the significance of future space exploration, and he never justified Apollo in terms of scientific payoffs
- Rather, he was a pragmatic national leader and politician who came to the judgment that the U.S. national interest required the United States to have a preeminent position in space capabilities and achievements and chose the lunar landing mission as a way of demonstrating those capabilities
- Was this a “good” judgment? In 1961-1963? In retrospect?



Remarks in San Antonio, November 21, 1963



NASA MOVES FORWARD

- The Soviet Union never responded to JFK's invitation to cooperate and new President Lyndon Johnson did not push the initiative
- The interim Gemini program, begun in 1962 and with ten flights in 1965 and 1966, developed rendezvous and docking capabilities essential to Apollo
- After November 1963, there was no thought given to cancelling or delaying the program, even after the Apollo 1 launch pad fire in January 1967 killed three astronauts. Apollo became a memorial to a fallen young president.



Gemini 6 and Gemini 7 rendezvous



Apollo 1 spacecraft

ANOTHER CRITICAL DECISION

APOLLO 8 TO ORBIT THE MOON

- By mid-1968, NASA was ready to resume Apollo flights, but the lunar landing module was behind schedule
- In a bold decision, NASA decided to send the Apollo 8 mission to orbit the Moon with a crew aboard but without a landing module attached
- Possibility of a Soviet circumlunar flight being first to reach the Moon influenced NASA decision
- December 1968 Apollo 8 mission was totally successful; the path to a lunar landing was open.



JULY 16, 1969 – APOLLO 11 LAUNCH



First and Second Men on the Moon



WHAT NEXT?

- President Lyndon B. Johnson explicitly left decisions on what should follow Apollo to his successor.
- That successor turned out to be Richard Nixon. To make a “definitive recommendation” on post-Apollo space activities, Nixon in February 1969 created a Space Task Group chaired by Vice President Spiro Agnew.
- The Space Task Group met between March-September 1969; it was captured by the expansive views of new NASA Administrator Tom Paine, backed by Agnew.
- Paine decided to use a human missions to Mars in the 1980s as the justification for the space station and space shuttle that NASA wanted to develop in the 1970s.



President Richard Nixon, NASA Administrator Tom Paine, and Vice President Spiro Agnew

RECOMMENDATIONS OF SPACE TASK GROUP

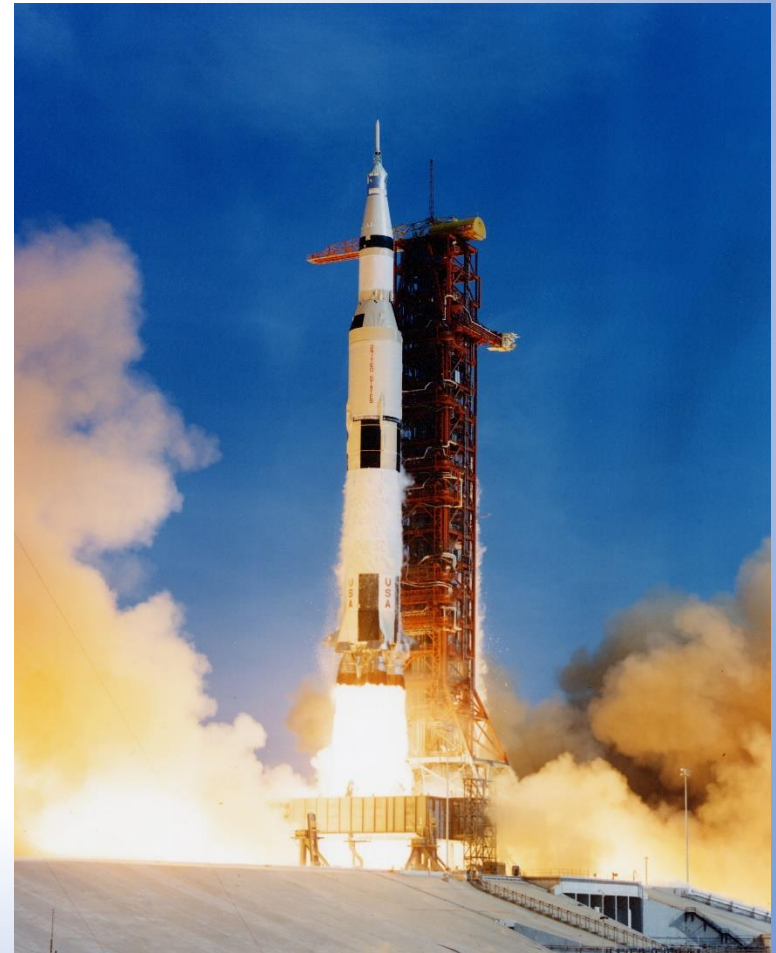
COMPARITIVE PROGRAM ACCOMPLISHMENTS

MILESTONES	MAXIMUM PACE	PROGRAM I	II, III	LOW LEVEL
<u>MANNED SYSTEMS</u>				
Space Station (Earth Orbit)	1975	1976	1977	-
50-Man Space Base (Earth Orbit)	1980	1980	1984	-
100-Man Space Base (Earth Orbit)	1985	1985	1989	-
Lunar Orbiting Station	1976	1978	1981	-
Lunar Surface Base	1978	1980	1983	-
Initial Mars Expedition	1981	1983	II - 1986 III - Open	-
<u>Space Transportation System</u>				
Earth-to-Orbit	1975	1976	1977	-
Nuclear Orbit Transfer Stage	1978	1978	1981	-
Space Tug	1976	1978	1981	-
<u>Scientific</u>				
Large Orbiting Observatory	1979	1979	1980	-
High Energy Astron. Capability	1973	1973	1981	1973
Out-of-Ecliptic Survey	1975	1975	1978	1975
Mars-High Resolution Mapping	1977	1977	1981	1977
Venus-Atmospheric Probes	1976	1976	Mid-80's	1976
Multiple Outer Planet "Tours"	1977-79	1977-79	1977-79	1977-79
Asteroid Belt Survey	1975	1975	1981	1975
<u>Applications</u>				
Earliest Oper. Earth Resource System	1975	1975	1976	1975
Demonstration of Direct Broadcast	1978	1978	Mid-80's	1978
Demonstration of Navigation/Traffic Control	1974	1974	1976	1974

Table 1

THE END OF HUMAN EXPLORATION

- The Space Task Group report was dead on arrival. Nixon and his associates had no intention of supporting an ambitious and expensive post-Apollo program.
- During budget negotiations in December 1969, NASA Administrator Paine agreed to defer production of additional Saturn V boosters in order to get early funding for the new projects NASA desired, especially a large space station and space shuttle.
- Paine's sacrifice of the Saturn V was a recognition of the already clear Nixon decision that Apollo would not be quickly followed by another project aimed at deep space exploration
- With this decision, the U.S. gave up the heavy lift exploratory capability that was a necessary element of a continuing program of human space exploration. It has taken fifty years to regain that capability.



TRUNCATING PROJECT APOLLO

- Original plan called for lunar landing missions through Apollo 20.
- After the successes of Apollo 11 and Apollo 12, Apollo 20 was canceled in January 1970. Its Saturn V booster was assigned to the Skylab space station program
- A combination of the constrained future budget outlook plus concern by people like NASA's Robert Gilruth and George Low about risks of additional Apollo flights -- especially after Apollo 13 in April 1970 -- led NASA's leaders in September 1970 to cancel Apollo 15 and Apollo 19 and renumber the three remaining missions Apollo 15, 16, and 17.
- This cut short Project Apollo; there would be no more missions to the Moon after Apollo 17.

RICHARD NIXON AND THE MOON

- December 1969 – After Apollo 11 and 12, Nixon says that he “did not see the need to go to the moon six more times.”
- April 1970 – Nixon becomes emotionally involved in the fate of Apollo 13 crew and worries about risk of future Apollo missions.
- December 1970 – Nixon proposes canceling Apollo 17. In May 1971, he adds Apollo 16 as a candidate for cancellation.
- Nixon, November 24, 1971 – “I don't like the feeling of space shots between now and the [November 1972] election . . . I just don't feel as if the shots are a big deal at this time. There is also the risk that you could have another Apollo 13. . . . That would be the worst thing we could have. We are just not going to do it. There will not be any launches between now and the election, period.”



December 1969



September 1971

Apollo 17 – December 14, 1972

- Apollo 17 commander Gene Cernan:

“As I take man's last step from the surface, back home for some time to come . . . I'd like to just say what I believe history will record: that America's challenge of today has forged man's destiny of tomorrow. And, as we leave the Moon at Taurus-Littrow, we leave as we came and, God willing, as we shall return, with peace and hope for all mankind.”

- As Apollo 17 heads back to Earth, Richard Nixon issues a statement, saying “this may be the last time in this century that men will walk on the Moon.”



The ascent stage of the lunar module *Challenger* leaves the Moon's Taurus-Littrow valley on December 14, 1972

THE KENNEDY SPACE LEGACY

- Of course, Apollo
- Link between space achievement and U.S. global leadership
- Gap between “real reasons” for going to the Moon and Kennedy’s public rhetoric
- Beating the Russians to the Moon goal resulted in expensive hardware not easily applicable to any space mission other than continued exploration
- Apollo mobilization created a space-industrial-Congressional complex centered on human spaceflight and large engineering projects
- Even though it was not sustainable, Apollo created an image of what a successful space program looks like – achieving a difficult goal on a challenging schedule

APOLLO AND U.S. POWER & PRIDE

- Apollo landings came at the end of a horrible decade in U.S. history. They provided a positive antidote to the events of the 1960s
- Veteran diplomat U. Alexis Johnson – “There is no question that the success of Apollo 11 mission did more to bolster prestige abroad than any single event since the termination of the Pacific War in 1945.”
- The image of a U.S. astronaut and the American flag on the lunar surface remains a potent symbol of American achievement



Apollo 11 astronauts welcomed in Mexico City

WHAT WERE THE IMPACTS OF APOLLO?

- In terms of goals set by JFK
- In terms of impact on post-Apollo space development
- In terms of impact on U.S. technology, education, and economy
- In terms of human history

“The politics of the moment had become linked to the dreams of centuries”

IF WE CAN PUT A MAN ON THE MOON, WHY CAN'T WE . . . ?

- The belief persists that Apollo can serve as a model for other long-term, large-scale undertakings
- But the conditions that made Apollo possible are difficult, perhaps impossible, to reproduce
 - The objective sought was technologically feasible
 - There existed a dramatic “occasion for decision”
 - Government leadership and public were supportive of undertaking positive government actions with high short-term costs and long-term benefits
- Perhaps the Apollo experience was unique, and its most relevant lesson is that governments can achieve great things under the right set of circumstances, but that those circumstances will vary over time and objective sought

WILL ANYTHING LIKE THE “KENNEDY MOMENT” COME AGAIN?

